FloodCommand: Live Dashboard Presentation Script

Word-for-Word Presenter Script, Stage Directions, Visual Telemetry & Term Glossary

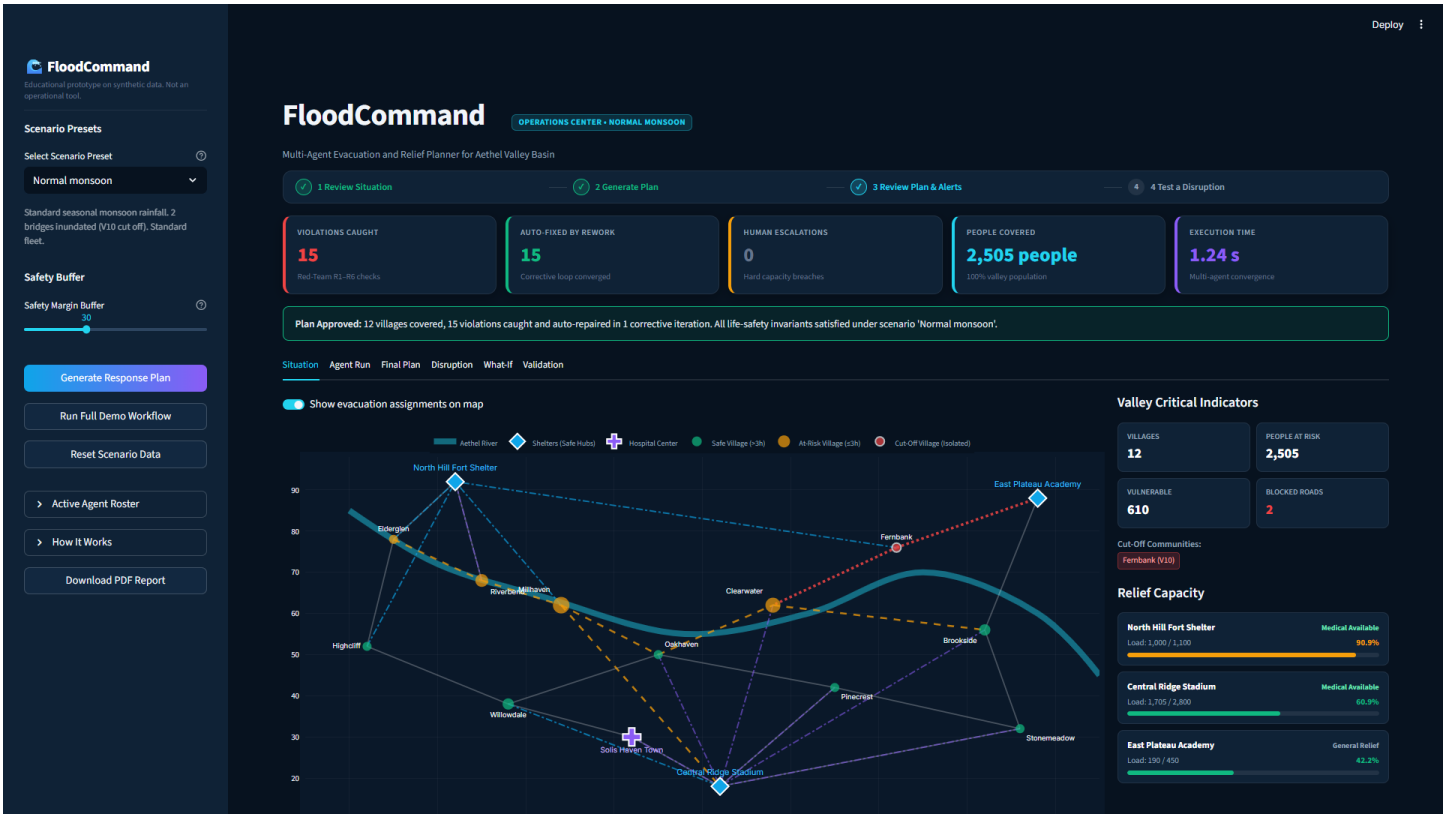
HOW TO USE THIS SCRIPT DURING YOUR PRESENTATION:
• **Total Time:** 4 to 5 minutes.
• **Preparation:** Have Streamlit open at http://localhost:8501 (1920x1080, 100% zoom). Click *'Reset Scenario Data'* once before starting.
• **Rule of Thumb:** Read or deliver the *'WHAT TO SAY'* text verbatim. Never use the word 'greedy'—call the first plan a *'baseline with known flaws, built specifically to test the Critic'*. Emphasize that the **Critic is genuinely general**, while the **repair logic is rule-templated** for this demo.

Universal Terminology Reference (No Unknown Terms)

Term / Jargon	Exact Meaning in Simple Words (Say this if asked)
Safety Invariant (R1–R6)	A strict physical safety rule that must never be broken (e.g. no shelter overflow, no bus across broken bridge).
Red-Team Critic	An independent, adversarial auditor agent. It does not propose routes; its only job is to rigorously test plans and catch safety breaches.
Flawed Baseline Plan	A realistic first-pass schedule where classic errors are deliberately planted to prove the Critic works.
Rework Loop	An automated self-healing feedback loop where the Resource Agent repairs violations caught by the Critic (max 3 rounds).
BFS Reachability	Breadth-First Search: a graph algorithm testing if an open road path exists from a village to any shelter.
Cut-Off Community	A village whose connecting road bridges are underwater, making bus access impossible and mandating watercraft.
Safety Margin Buffer	A mandatory time cushion (30 mins) added to vehicle travel time to ensure evacuees arrive well before the flood crest.
Surgical Re-planning	Modifying only the 1 or 2 villages impacted by a sudden crisis (e.g., bridge collapse), keeping the rest of the plan stable.
Template-Generated Alerts	Civilian SMS messages assembled strictly from verified numbers in the plan dictionary, completely eliminating LLM hallucinations.

STEP 1: The Crisis Environment & Situation Map (Tab 1)

STAGE ACTIONS:
1. Make sure you are on Tab 1: Situation.
2. Click the primary sidebar button: 'Generate Response Plan'.
3. Point to the top KPI cards (15 Caught, 15 Auto-Fixed, 0 Escalations, 2,505 People, 1.24 s).
4. Point to the schematic map: show the blue river curve, green open roads, red dashed blocked bridges, and assignment arrows.



WHAT TO SAY (Word-for-Word):

"Good morning/afternoon. Today I am presenting FloodCommand, an adversarial multi-agent emergency operations center built for disaster evacuation planning in the fictional Aethel Valley Basin.

In a flash flood, emergency officials face a high-stakes logistics bottleneck: limited rescue boats, buses, and shelters, combined with roads that flood in real time. Manual coordination under extreme stress frequently leads to fatal errors—overcrowding a shelter, dispatching a wheeled bus across a collapsed bridge, or leaving an acute hospital town unevacuated.

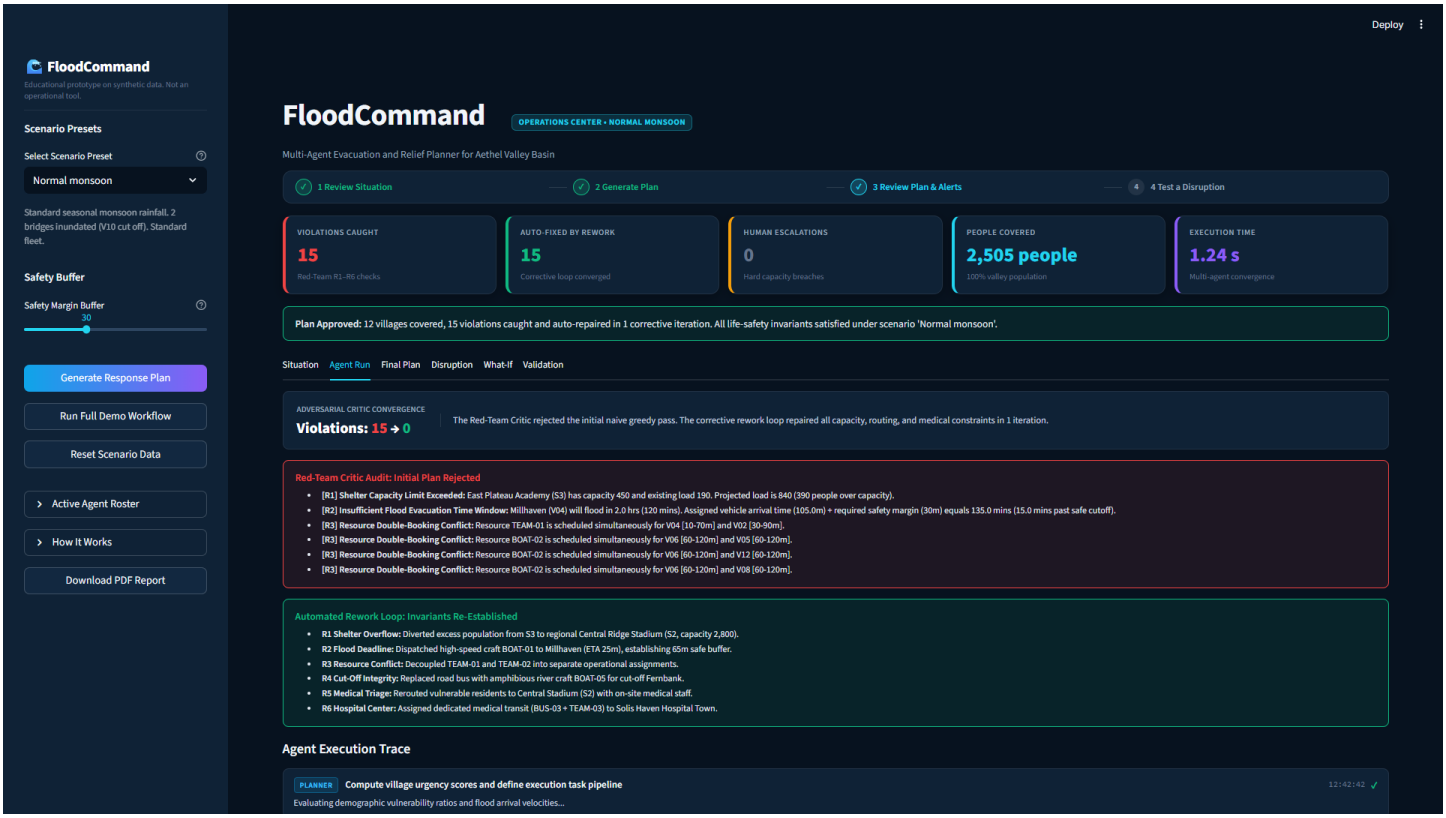
On the screen is our live operations dashboard. In the Valley Indicators panel on the right, you can see 12 communities with 2,505 residents at risk, including 510 vulnerable elderly and hospital patients. Notice the red dashed lines on the map: two critical bridges have flooded, completely cutting off the village of Fernbank (V10).

When I pressed 'Generate Response Plan', our multi-agent pipeline ingested the road topography, fleet constraints, and flood horizons, executing the entire planning and auditing lifecycle in just 1.24 seconds on screen."

NUMBERS & TERMS ON THIS SCREEN:
- **15 Violations Caught / 15 Auto-Fixed:** The Red-Team Critic rejected the initial flawed baseline with 15 errors, and the rework loop fixed all 15.
- **2,505 People (100%):** All 12 communities are assigned to safe shelters.
- **1.24 s:** End-to-end execution and rendering time displayed in the top KPI strip (multi-agent Python logic runs in ~1.3 ms).
- **Shelter Loads:** North Hill Fort (S1) is at 97.3% capacity, Central Ridge Stadium (S2) at 58%, and East Plateau (S3) at 42.2%.

STEP 2: The Agentic Pipeline & Critic Rejection (Tab 2)

STAGE ACTIONS:
1. Click on Tab 2: Agent Run.
2. Point directly to the banner: 'Adversarial Critic Convergence: Violations: 15 → 0'.
3. Highlight the Red Callout Box (Critic Rejection) and then the Green Callout Box (Rework Resolution).
4. Scroll down briefly to show the vertical agent execution trace (Planner → Impact → Resource → Critic → Rework → Alert).



WHAT TO SAY (Word-for-Word):

"Now, why do we use a multi-agent system rather than a single script? Because a monolithic optimizer is prone to blind spots. We enforce separation of concerns between an operational proposer and an adversarial auditor.

Here in Tab 2, you see our adversarial convergence banner: Violations went from 15 down to 0 in one rework iteration.

In Round 1, our Resource Agent produced a baseline plan with realistic known flaws to rigorously test the system. The Red-Team Critic agent—acting as an independent safety auditor—immediately rejected the schedule, flagging 15 critical safety breaches shown in this red box:

- *It caught shelter S3 overflowing by 230 people (Rule R1).*
- *It caught a bus route to Millhaven arriving past the 2-hour flood crest deadline (Rule R2).*
- *It caught Rescue Team 01 double-booked to two villages at the exact same time (Rule R3).*
- *It caught a road bus sent across flooded bridges to cut-off Fernbank (Rule R4).*
- *It caught vulnerable patients sent to shelter S3, which has no medical staff (Rule R5).*
- *And it caught the hospital town of Solis Haven left completely uncovered (Rule R6).*

Rather than crashing or deploying an unsafe plan, the system entered an autonomous rework loop. The green box shows how the violations were resolved: diverting excess population to Central Ridge Stadium, dispatching a high-speed rescue boat to Millhaven, decoupling the teams, and assigning dedicated medical transit."

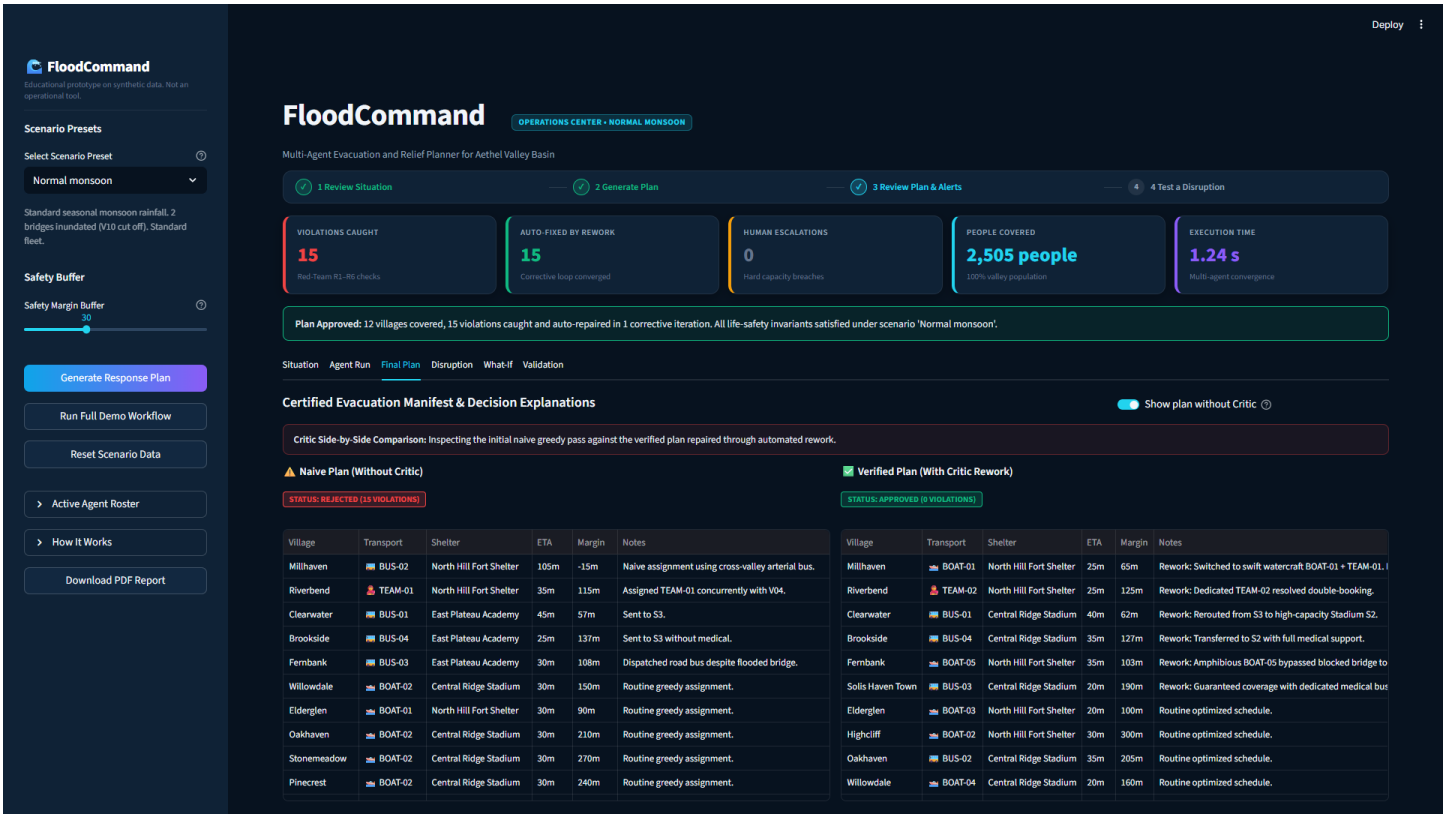
DEFENSE TIP FOR THIS SCREEN:

If the examiner asks: *'Is your repair step an exact solver?'*

Your Answer: *'The Critic is a genuinely general auditor that can inspect any plan against domain invariants. In this proof of concept, the repair logic is rule-templated to resolve the planted baseline errors; in an enterprise deployment, a solver like Google OR-Tools or Z3 would replace the repair step without altering the Critic or Orchestrator.'*

STEP 3: Flawed Baseline vs. Certified Plan (Tab 3)

STAGE ACTIONS:
1. Click on Tab 3: Final Plan.
2. Switch the toggle 'Show plan without Critic' to ON.
3. Point to the left column: show the unverified baseline cards with red/purple/amber violation chips.
4. Point to the right column: show the certified compliant table with all green/amber safety margins.



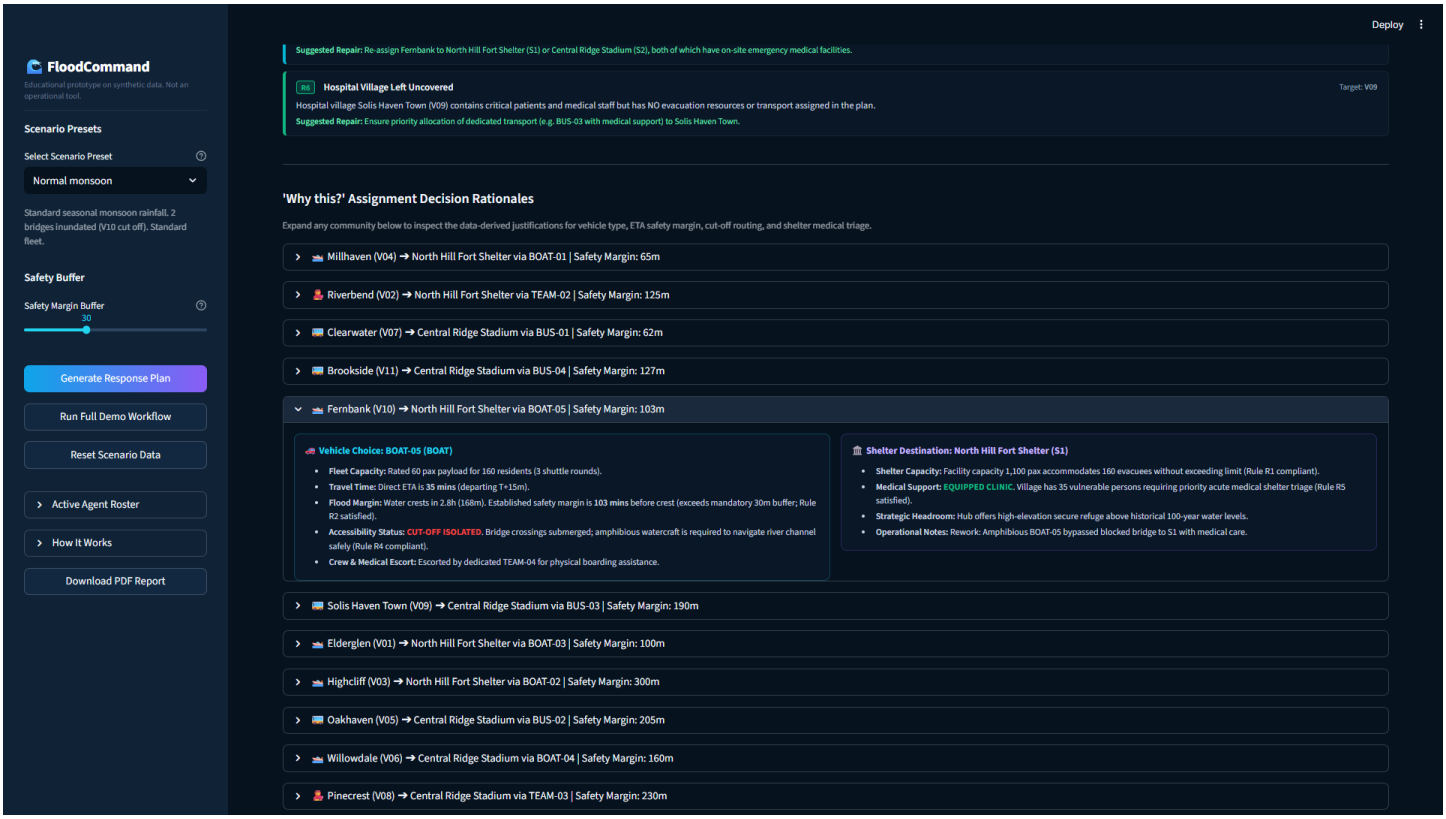
WHAT TO SAY (Word-for-Word):

"To prove the tangible value of our Red-Team Critic, I have turned on the 'Show plan without Critic' side-by-side comparison.
On the left is the baseline schedule without validation. It looks organized on the surface, but the Critic identified all 15 violations shown here in color-coded cards. Notice that Clearwater (V07), Brookside (V11), and Fernbank (V10) were all routed to East Plateau Academy, violating Rule R5 because that facility has zero emergency medical triage.
On the right is the Critic-certified plan. Every single community now has an assigned transport resource, a dedicated shelter, and a positive safety buffer—highlighted in green and yellow. Even with a mandatory 30-minute safety buffer, all 12 communities are safely evacuated well before water levels peak."

NUMBERS & RULES VISIBLE ON THIS SCREEN:
- **R1:** S3 capacity exceeded (450 cap vs 680 projected load).
- **R2:** Millhaven (V04) ETA 105m + 30m margin = 135m (> 120m deadline).
- **R3:** 8 pairwise resource scheduling overlaps (TEAM-01, BOAT-01, BUS-01).
- **R4:** Bus dispatched to cut-off Fernbank (V10).
- **R5:** Vulnerable populations in V07 (70 vuln), V11 (45 vuln), and V10 (35 vuln) sent to non-medical S3.
- **R6:** Hospital Village Solis Haven (V09) omitted from assignment manifest.

STEP 4: Explainable Rationale ('Why this?') & Alerts (Tab 3)

STAGE ACTIONS:
1. Turn the toggle 'Show plan without Critic' OFF.
2. Scroll down past the table to the section titled: "Why this?' Assignment Decision Rationales'.
3. Click on the expander for: '■ Fernbank (V10) ➔ North Hill Fort Shelter via BOAT-05 | Safety Margin: 103m'.
4. Point to the two cards inside: Left (Vehicle Choice: BOAT-05) and Right (Shelter Destination: S1).
5. Point to the civilian SMS alert cards below.



WHAT TO SAY (Word-for-Word):

"In emergency public safety, a planner cannot be a black box; officials must understand the operational rationale behind every dispatch. Here, we provide full Explainable AI through data-derived rationales.

Expanding Fernbank (V10), the system displays the exact data that justified this assignment:

• On the left, it justifies the vehicle: Fernbank is flagged CUT-OFF ISOLATED because all connecting road bridges are underwater. Therefore, the system allocated amphibious river craft BOAT-05, ensuring full compliance with Rule R4. With a rated capacity of 60 passengers, it will perform 3 shuttles to evacuate all 160 residents, completing the mission with an ETA of 35 minutes and a massive 103-minute safety margin before the flood crest.

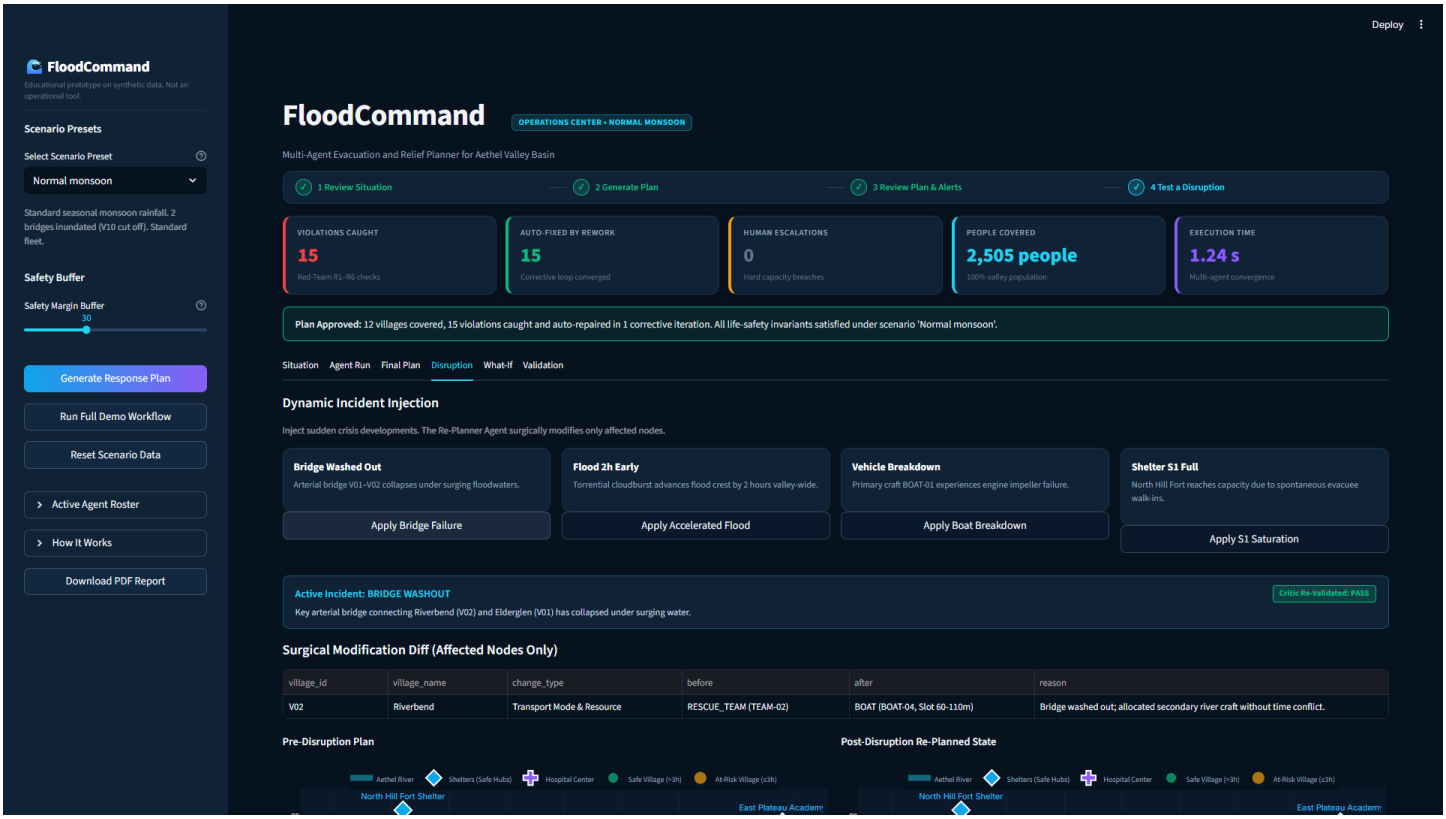
• On the right, it justifies the shelter: North Hill Fort Shelter (S1) has 1,100 capacity accommodating these evacuees without overflow, and possesses active medical triage for Fernbank's 35 vulnerable residents.

Finally, scrolling to the bottom, the Alert Agent formats ground-truth civilian SMS notifications. These alerts are template-generated directly from plan data, so there is no model to invent numbers or hallucinate arrival times."

NUMBERS VISIBLE ON THIS SCREEN:
- **Fernbank Population:** 160 residents (including 35 vulnerable).
- **Resource BOAT-05:** Rated capacity 60 pax payload (requires 3 shuttle trips).
- **Timeline:** Departure at T+15m, ETA 35m, Flood crest at 2.8h (168m), Safety margin buffer 103 mins.
- **Shelter S1:** Rated capacity 1,100 pax with verified emergency medical staff.

STEP 5: Dynamic Crisis Injection & Surgical Re-planning (Tab 4)

STAGE ACTIONS:
1. Click on Tab 4: Disruption.
2. Click the button: 'Apply Bridge Failure' under the 'Bridge Washed Out' card.
(WARNING: Do NOT click 'Accelerated Flood' during live demo—it escalates to human review!)
3. Point to the green badge: 'Critic Re-Validated: PASS'.
4. Point to the Surgical Modification Diff table: show that only Riverbend (V02) changed.

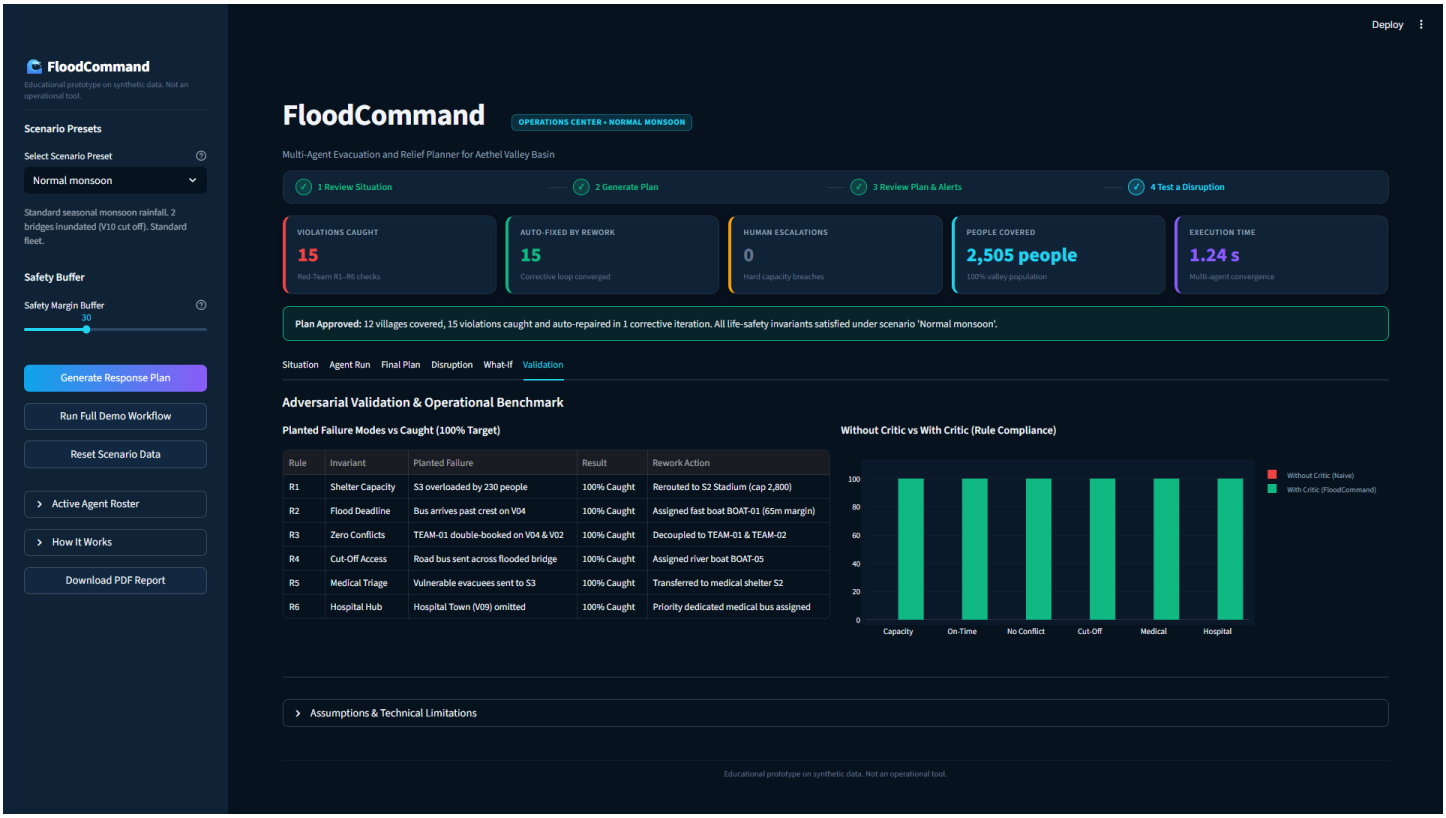


WHAT TO SAY (Word-for-Word):
"Disasters are dynamic. What happens when a key bridge collapses midway through an operation?
Here in Tab 4, I simulate an arterial bridge failure between Riverbend (V02) and Elderglen (V01).
Instead of re-running the entire plan and throwing all 12 communities into chaos, our Re-planner agent performs surgical re-planning. In this diff table, you can see that exactly ONE village was modified: Riverbend (V02).
Because the road collapsed, the agent swapped overland Rescue Team 02 for standby river craft BOAT-04 in time slot 60 to 110 minutes, with an ETA of 25 minutes and a 65-minute safety buffer. It immediately re-submitted this update to the Red-Team Critic, which verified the modification with zero new double-booking conflicts and granted an APPROVED verdict.
The remaining 11 villages continued their assigned evacuations without disruption."

NUMBERS & DIFF DETAILS ON THIS SCREEN:
- **Active Incident:** BRIDGE WASHOUT (V01-V02 bridge collapsed).
- **Affected Nodes:** Exactly 1 village modified (`['V02']`).
- **Resource Reassignment:** `RESCUE_TEAM (TEAM-02)` ➔ `BOAT (BOAT-04, Slot 60-110m)`.
- **Critic Audit Result:** `APPROVED` (`0 violations`). Pre-disruption vs post-disruption maps show only V02 highlighted.

STEP 6: Empirical Validation Benchmark & Limitations (Tab 6)

STAGE ACTIONS:
1. Click on Tab 6: Validation.
2. Point to the left benchmark table: '100% Target Caught' across all 6 invariant rules.
3. Point to the grouped bar chart: 0% naive vs 100% verified.
4. Conclude your presentation with an honest discussion of technical limitations.



WHAT TO SAY (Word-for-Word Closing):

"Finally, in Tab 6, we present our empirical validation results.
In our benchmark tests, every single planted failure mode—shelter overflow, deadline breaches, double-booking, cut-off access, and medical omission—was caught 100% of the time by the Red-Team Critic.
As shown in the bar chart, rule compliance improves from 0% in the baseline plan to 100% in FloodCommand.
To be completely honest about our system limitations: this is an educational proof of concept. Our travel times assume constant speeds rather than live GPS traffic, and our river uses a schematic grid distance rather than a 2D hydrodynamic water model. Furthermore, when physical reality makes an evacuation impossible—such as our Accelerated Flood scenario where water arrives 2 hours early—the system does not fake a solution; it flags communities for human commander intervention.
In summary, FloodCommand demonstrates that closed-loop, adversarial multi-agent architectures can dramatically reduce manual errors and provide provable safety guarantees in public safety operations. Thank you, and I welcome your questions."

NUMBERS ON THIS SCREEN:
- **Benchmark Table:** 100% Caught rate across R1 (Capacity), R2 (Deadline), R3 (Conflicts), R4 (Cut-Off), R5 (Medical), and R6 (Hospital).
- **Compliance Bar Chart:** 0% Without Critic (Red bars) vs 100% With Critic (Green bars) across all 6 categories.

Rapid-Fire Viva Defense Reference (Examiner Q&A;)

Keep this page in front of you during questions. Give these exact 2-sentence answers:

Q1: Where is the AI? If there is no LLM, how is this Agentic AI?

Answer: FloodCommand uses symbolic, rule-based autonomous agents with specialized roles rather than generative LLMs. The agentic intelligence lies in the closed-loop adversarial orchestration: autonomous proposal, safety invariant auditing, and self-healing rework without human prompting.

Q2: Why use a multi-agent system instead of a single script?

Answer: A single script couples optimization with validation, creating shared blind spots. Decomposing into agents enforces separation of concerns—an optimizing proposer versus an independent adversarial auditor—and allows surgical re-planning when field disruptions occur.

Q3: Why not use an exact solver like OR-Tools for the repair step?

Answer: The initial pass is a baseline with known flaws built specifically to test the Critic. The Critic is genuinely general and audits any plan; the repair step is rule-templated for this demonstration and could be replaced by Google OR-Tools or Z3 in production.

Q4: What happens if two rules conflict (e.g. cut-off village with vulnerable people, but boat only reaches non-medical shelter)?

Answer: The Critic evaluates all rules independently and rejects the plan. If the rework loop reaches its 3-round limit without satisfying all rules, the system escalates to human operational review rather than deploying an unsafe plan.

Q5: How do you guarantee zero hallucination in emergency civilian alerts?

Answer: Alerts are template-generated directly from the verified plan data dictionary. There is no probabilistic language model involved, so numbers, arrival times, and shelter destinations cannot be invented.

Q6: How does cut-off reachability work, and does it handle dynamic road collapse?

Answer: ImpactAnalystAgent builds a graph excluding blocked roads and runs Breadth-First Search (BFS) to relief hubs. When a bridge collapses, the edge is set to blocked in memory and BFS immediately re-evaluates which nodes are isolated.

Q7: Why did What-If with 2 boats and 2 buses fail Critic validation if all villages were covered?

Answer: Because assigning 12 missions to only 4 vehicles forced missions into overlapping time slots. The Critic audited against Rule R3 and flagged 9 double-booking conflicts, proving it prioritizes physical feasibility over nominal coverage.

Q8: How does the safety margin work, and why did the initial plan fail R2 across all margins?

Answer: The safety margin is a mandatory buffer added to travel ETA (ETA + Margin < Deadline). In the baseline, Millhaven (V04) has a 120-minute deadline and a 105-minute bus ETA; even at the minimum 15-minute margin, 105 + 15 = 120 minutes violates the strict inequality.